



ASCENDION

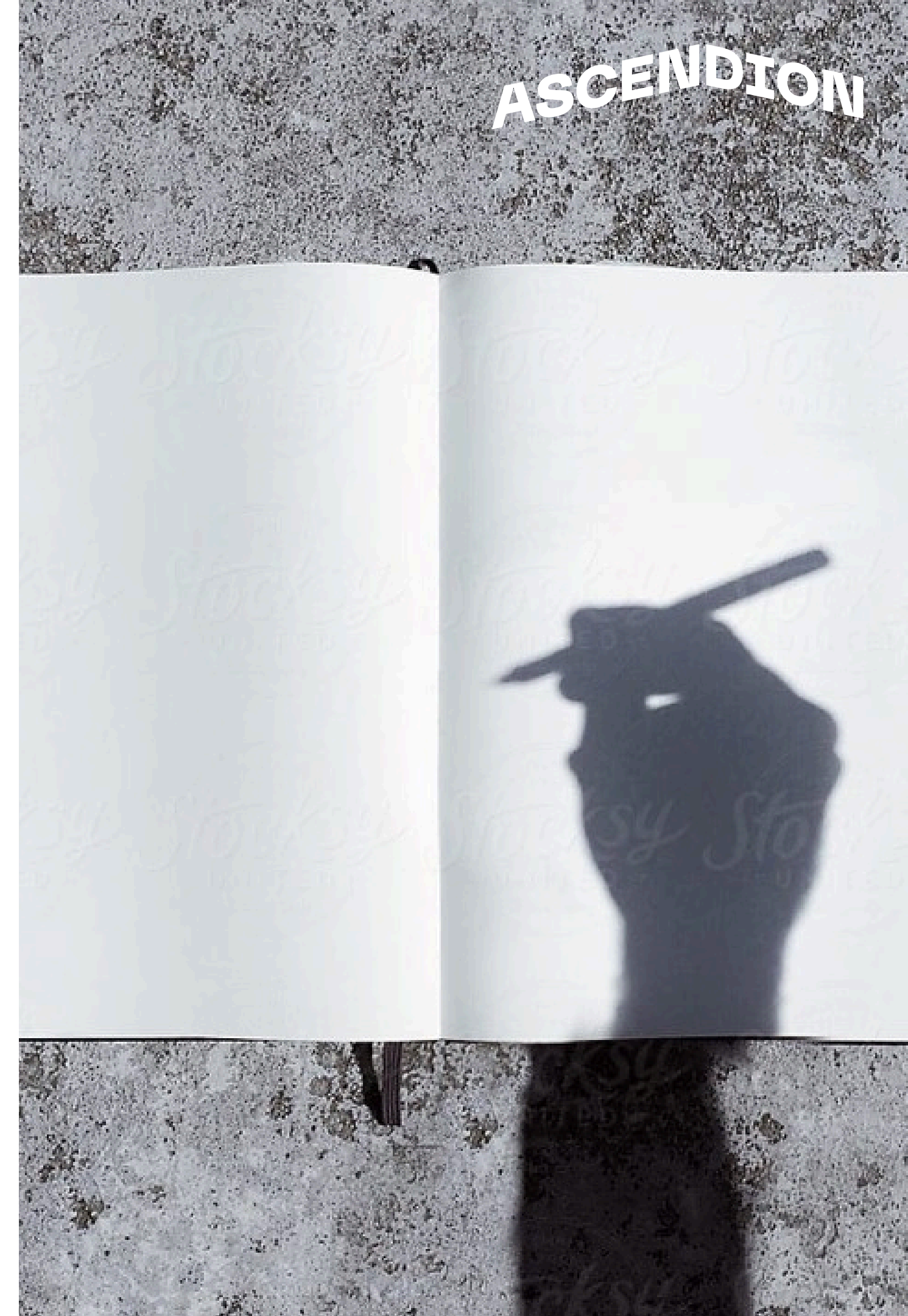
HEALTHCARE MANAGEMENT SYSTEM

Personal BACKGROUND

(Name, Qualification)

Name : PEDDINENI BAVITHA

Qualification : B.E. / B.Tech — Computer Science and Engineering, PES University



Key Takeaways/Learnings from the Program (HTD)

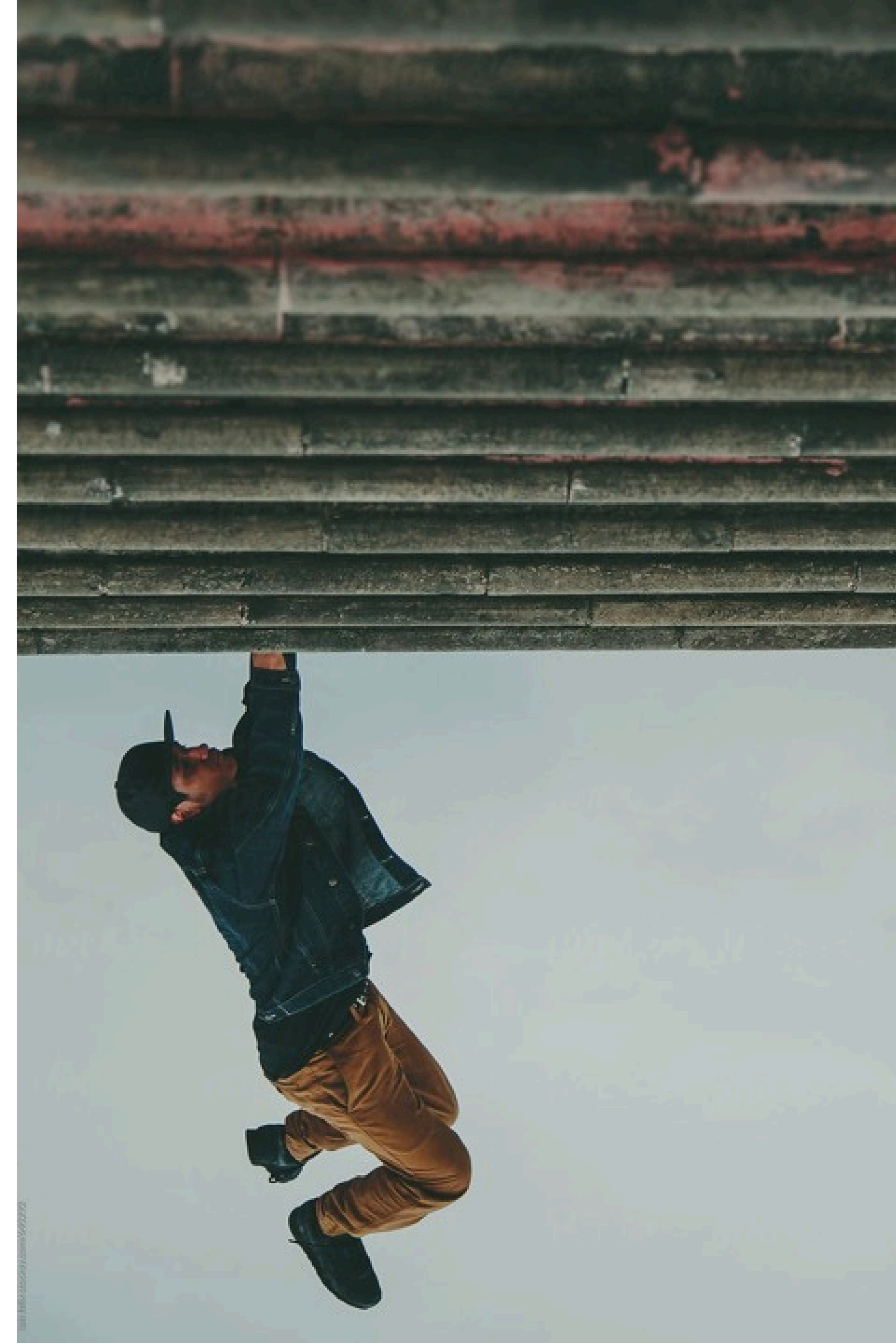
- Designed and deployed a microservices architecture using Spring Boot & hibernate.
- Implemented Eureka Service Registry and Spring Cloud API Gateway for routing.
- Built a React frontend with authentication, dashboards, and CRUD tables.
- Integrated Admin Service with secure login and logout mechanism.
- Learned best practices in REST API design, database modeling, and frontend-backend integration.



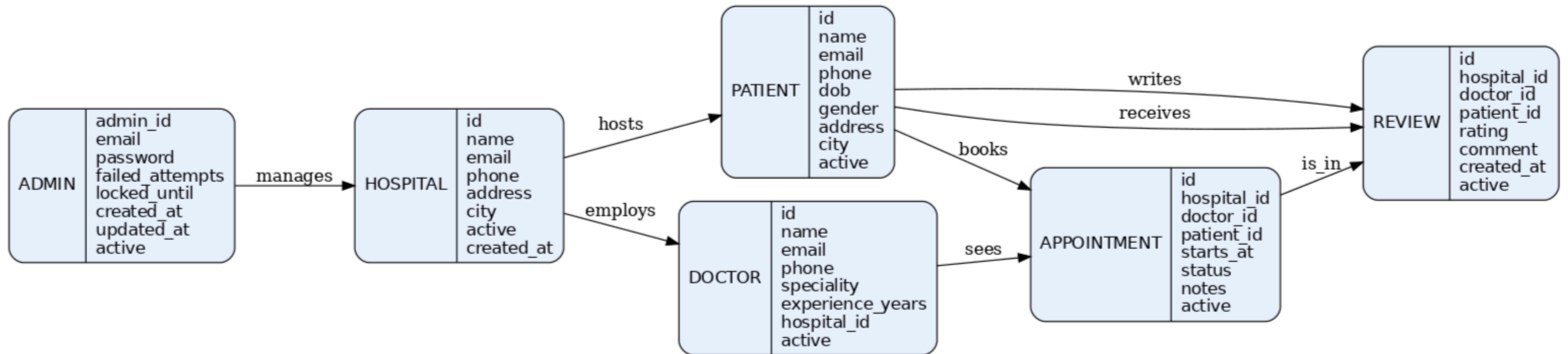
Problem Statement of the Capstone Project

Challenges in Healthcare Management System

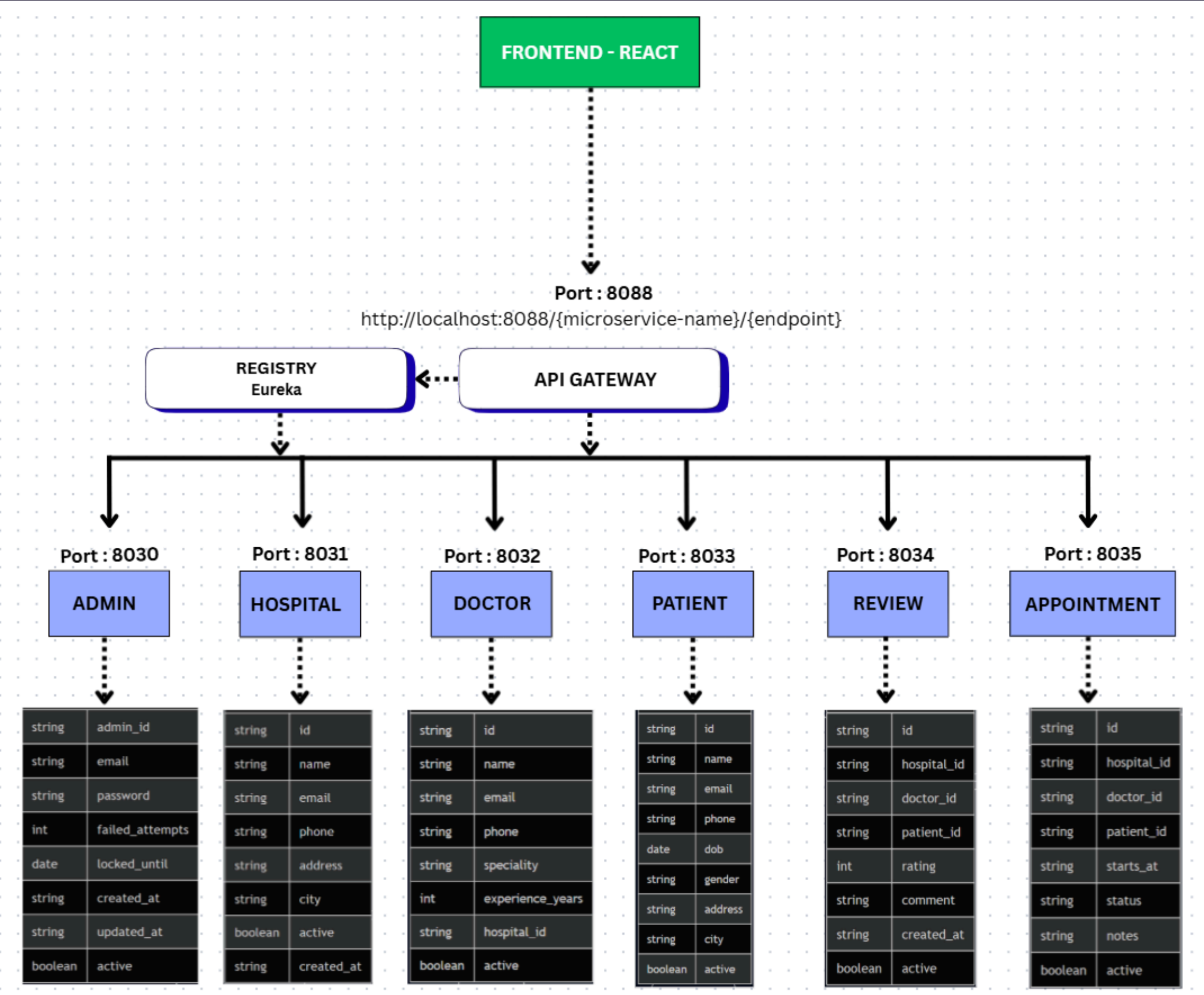
- Healthcare providers need a centralized platform to manage multiple entities:
- Hospitals, Doctors, Patients, Appointments, and Reviews.
- Manual systems are prone to inefficiency, duplication, and errors.
- A microservices-based solution enables scalability, modularity, and ease of maintenance.



ER DIAGRAM

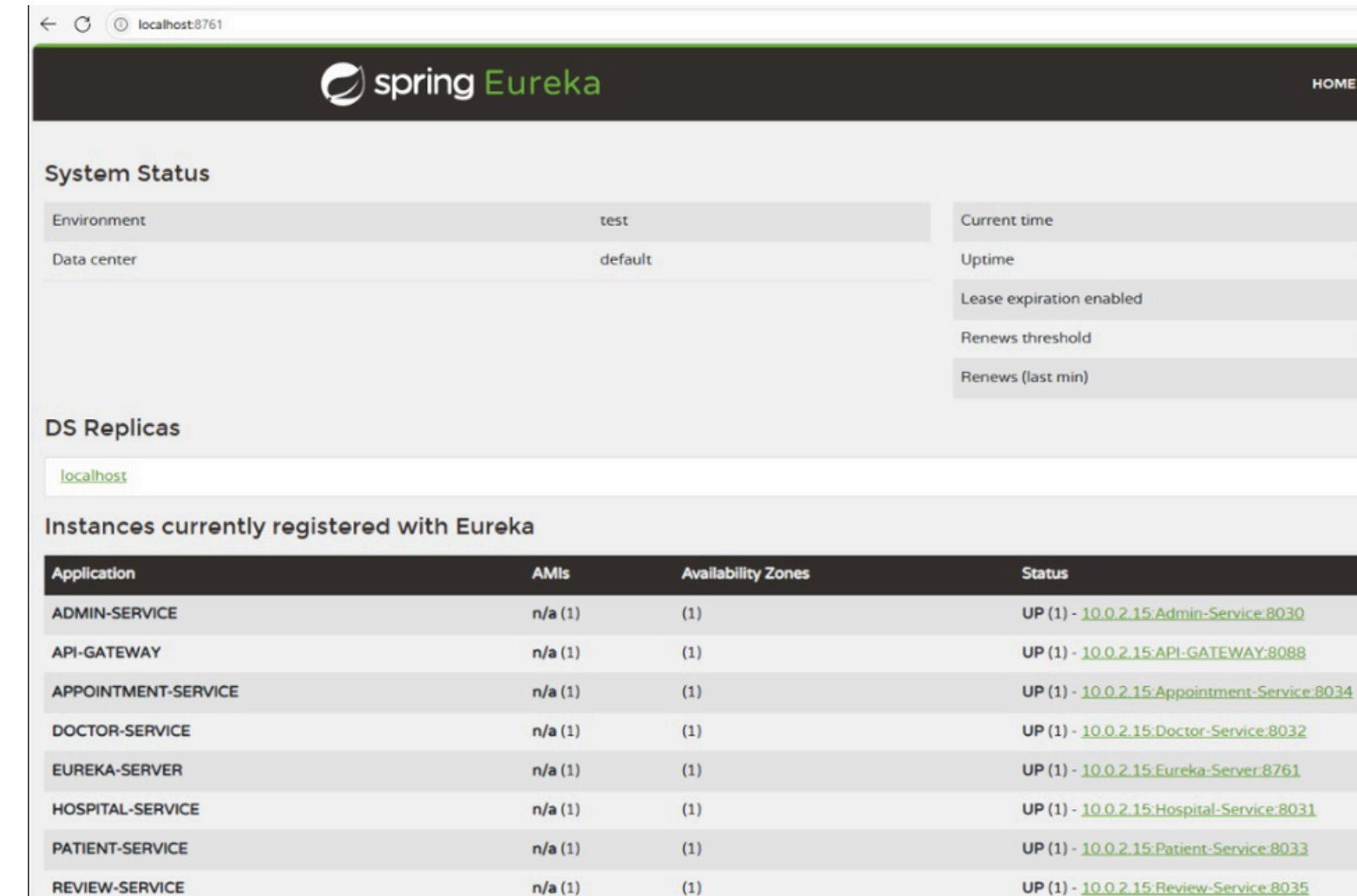


ARCHITECTURE



EUREKA SERVER

- **Central Registry:** Eureka Server acts as the service registry where all microservices register themselves.
- **Dynamic Discovery:** Instead of hardcoding IPs/ports, services look up other services through Eureka.
- Admin-Service, Doctor-Service, Patient-Service, Hospital-Service, Appointment-Service, Review-Service
- → These register with Eureka so they can find each other without manual configuration.
- API Gateway:
 - Routes all client requests to the correct microservice using Eureka's registry.
- Eureka Server :
 - → Runs on port 8761 and maintains the registry list (as seen in the screenshot).
- Status "UP"
- → Confirms each service is healthy and available for communication.



The screenshot shows the Spring Eureka web interface running on localhost:8761. The page has a dark header with the 'spring Eureka' logo and a 'HOME' link. The main content area is divided into two columns. The left column contains 'System Status' and 'DS Replicas'. The right column contains system metrics like 'Current time', 'Uptime', 'Lease expiration enabled', 'Renews threshold', and 'Renews (last min)'. Below these, there is a section 'Instances currently registered with Eureka' which contains a table listing various services and their status.

Application	AMIs	Availability Zones	Status
ADMIN-SERVICE	n/a (1)	(1)	UP (1) - 10.0.2.15:Admin-Service:8030
API-GATEWAY	n/a (1)	(1)	UP (1) - 10.0.2.15:API-GATEWAY:8088
APPOINTMENT-SERVICE	n/a (1)	(1)	UP (1) - 10.0.2.15:Appointment-Service:8034
DOCTOR-SERVICE	n/a (1)	(1)	UP (1) - 10.0.2.15:Doctor-Service:8032
EUREKA-SERVER	n/a (1)	(1)	UP (1) - 10.0.2.15:Eureka-Server:8761
HOSPITAL-SERVICE	n/a (1)	(1)	UP (1) - 10.0.2.15:Hospital-Service:8031
PATIENT-SERVICE	n/a (1)	(1)	UP (1) - 10.0.2.15:Patient-Service:8033
REVIEW-SERVICE	n/a (1)	(1)	UP (1) - 10.0.2.15:Review-Service:8035

POSTMAN QUERIES

workspace

NewImport

+

Search collections

HCMS (Admin_Service)

GET Login

GET Register

GET GetAllAdmins

GET GetById

GET DeleteById

HCMS (Appointment_Service)

GET AddAppointment

GET New Request

HCMS (Doctor_Service)

GET AddDoctor

GET GetAllDoctors

GET GetDoctorById

GET DeleteById

GET Search

HCMS (Hospital_Service)

GET AddHospital

GET GetAllHospitals

GET GetByID

GET Search

GET Delete

HCMS (Patient_Service)

GET AddPatients

GET New Request

HCMS (Review_Service)

Virtual_Learning_System

BDEL DeletPOST AddGET GetAGET GetBGET SearDEL DeleHCMS

HCMS (Appointment_Service) / AddAppointment

POST

http://localhost:8034/api/appointments

Params

Authorization

Headers (8)

Body

Scripts

Tests

Settings

none

form-data

x-www-form-urlencoded

raw

binary

GraphQL

JSON

1

2

3

4

5

6

7

8

{

"patientId": "P0001",

"doctorId": "D0001",

"whenAt": "2025-09-05T15:30:00",

"reason": "Follow-up",

"status": "SCHEDULED"

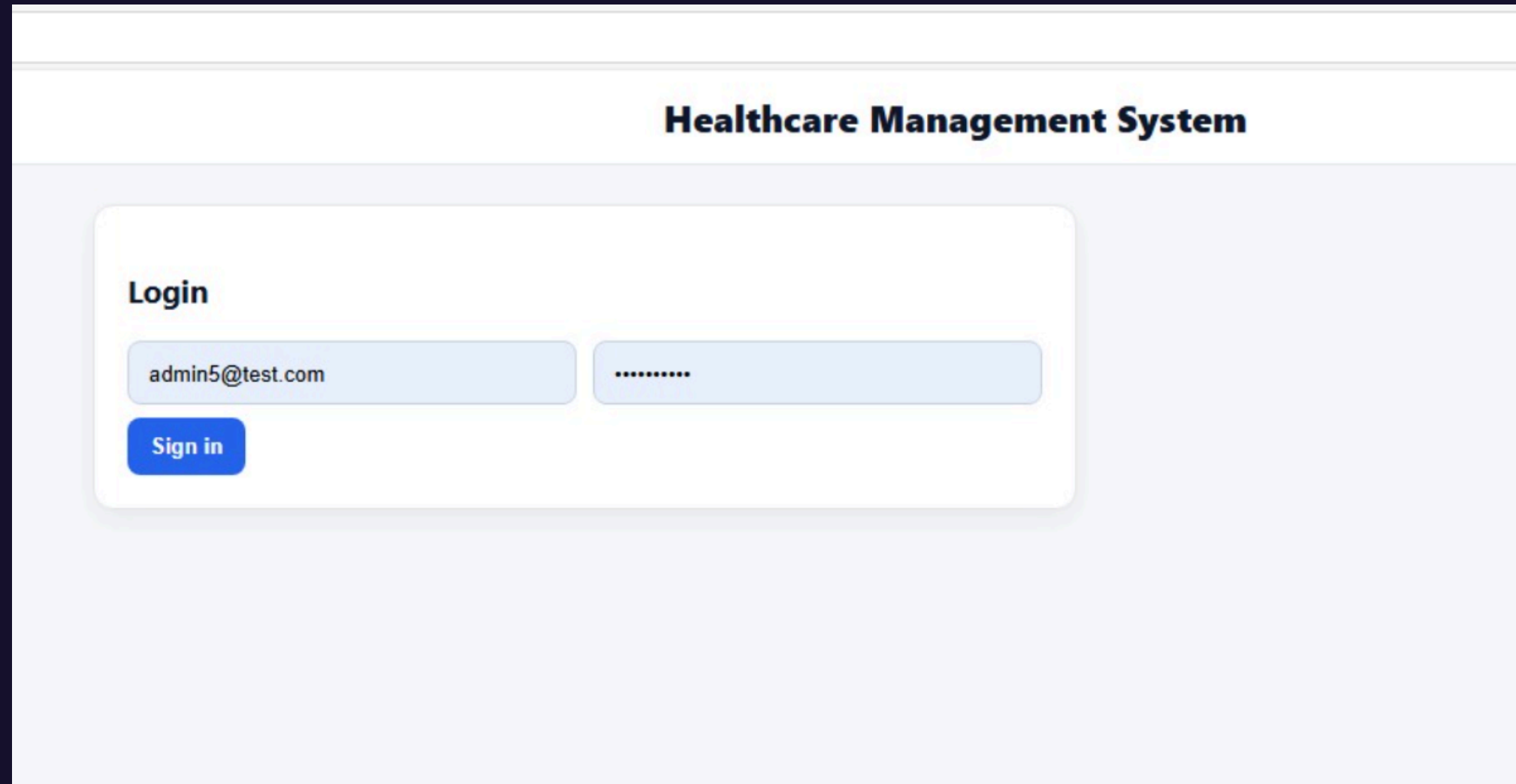
}

Response

History



LOGIN

A screenshot of a web application's login page. The page has a white background with a light gray header bar. The header bar contains the text "Healthcare Management System" in a bold, black, sans-serif font. Below the header, there is a white rectangular box with rounded corners and a subtle drop shadow. Inside this box, the word "Login" is written in a bold, black font. Below "Login", there are two input fields: the first contains the email address "admin5@test.com" and the second contains a series of dots representing a password. Below these fields is a blue button with the text "Sign in" in white.

- Credentials entered here are validated against the ADMIN table in the database.
- Only admins with a valid email and password can successfully log in.
- If credentials match → login is granted, else → access is denied.
- Ensures role-based access control (only authorized admins manage the system).

LOGIN – 30 MINUTE LOCK

Healthcare Management System

Login

Sign in

Account locked. Try again later.

```
private boolean doctorExists(String doctorId) {  
    String sql = "SELECT COUNT(1) FROM Doctor WHERE DoctorId = :id";  
    Query q = entityManager.createNativeQuery(sql).setParameter("id", doctorId);  
    Number n = (Number) q.getSingleResult();  
    return n != null && n.longValue() > 0;  
}
```

- If an admin enters incorrect credentials 3 times consecutively, the account is automatically locked.
- During the lockout period, the admin cannot log in even with the correct password.
- The system uses the fields failed_attempts and locked_until in the Admin table to enforce this rule.
- After 30 minutes, the lock automatically expires, and the admin can attempt login again.

DASHBOARD



Healthcare Management System

Hospitals

Name *	Email	Phone
Address	City	
Add		

ID	Name	Email	Phone	Address	City	Actions
H0001	CityCare Hospital	citycare@example.com	9876543210	123 MG Road	Bangalore	Edit Delete
H0002	Apollo Hospitals	apollo@example.com	0801234567	Bannerghatta Road	Bangalore	Edit Delete
H0003	Fortis Healthcare	fortis@example.com	0809876543	Cunningham Road	Chennai	Edit Delete

Healthcare Management System

Doctors

Name *	Email	Phone
Specialization	Hospital ID *	
Add		

ID	Name	Email	Spec	Hospital	Actions
D0001	Dr. John Smit	john.smith@hospital.com	Cardiology	H0001	Edit Delete
D0002	Dr.Smith	john.smi@hospital.com	Cardiology	H0002	Edit Delete
D0003	Dr. Anita Desai	anita.desai@example.com	Dermatology	H0001	Edit Delete

Healthcare Management System

Patients

Name *	Email	Phone
Gender	Blood Group	Doctor ID *
Address	City	
Add		

ID	Name	Email	Phone	Gender	BG	Doctor	Actions
P0001	Alice Johnson	alice.johnson@demo.com	9876543219	Female	B+	D0002	Edit Delete
P0002	Bavitha p	bavithapeddineni@gmail.com	09148864782	female	A+	D0001	Edit Delete
P0003	Praneesh	peddineni@gmail.com	9060083237	male	A+	D0002	Edit Delete

Healthcare Management System

Reviews

Hospital ID *	5	mm/dd/yyyy --:-- --
Comment		
Add Review		

ID	Hospital	Rating	Comment	Created	Actions
R0001	H0001	5	Very good staff and clean environment.	2025-09-03T10:15:00	Delete
R0002	H0003	5	highly skilled doctors	2025-09-03T10:15:00	Delete
R0003	H0001	5	Great team!	2025-09-03T15:30:00	Delete
R0004	H0001	5	Great team!	2025-09-03T15:30:00	Delete
R0005	H0001	5	Great teammmm!	2025-09-03T15:30:00	Delete

Healthcare Management System

Appointments

Patient ID *	Doctor ID	mm/dd/yyyy --:-- --
Reason	SCHEDULED	
Add		

ID	Patient	Doctor	When	Status	Reason	Actions
A0001	P0001	D0001	2025-09-05T15:30:00	SCHEDULED	Follow-up	Edit Delete

Conclusion

- Successfully developed a full-stack Healthcare Management System using Microservices, React, and Spring Boot.
- Implemented secure authentication with admin login & account lock mechanism.
- Built modular microservices (Hospitals, Doctors, Patients, Appointments, Reviews) with independent scalability.
- Integrated Eureka Service Registry and API Gateway for dynamic routing and service discovery.
- Designed a React frontend dashboard with professional UI and CRUD tables.

BENEFITS

- Centralized Management
- Improved Efficiency
- Scalability
- Secure Access
- Better Patient Experience: Streamlined appointments and feedback system improve patient satisfaction.

ASCENDION

**Thank
You**